The project is divided into 7 phases. You must deliver a report for each phase.

Phase 1: The "Dirty Data" Audit (Data Cleaning)

This is your first test. The engineering team warns that the sensors might be glitching.

The Client asks:

1. **The "Ghost" Rides:** "We are seeing trips with trip_distance = 0 but the meter was still running. How many of these exist? Are they valid (e.g., stuck in traffic) or system errors? Please analyse and handle them."
2. **The Negative Money Problem:** "Our accounting team found rows with **negative** fare_amount and total_amount. We can't have negative revenue. Identify these rows. Is it a specific payment_type causing this? Remove or fix them."
3. **The Silent Passenger:** "We have trips with passenger_count = 0. Is this possible? (maybe package delivery?). If it's less than 1% of data, remove it. If it's more, keep it and explain why."
4. **The Time Travelers:** "Check for trips where the dropoff_datetime is *before* the pickup_datetime. These are obvious errors."

Phase 2: Operational Intelligence (EDA & SQL)

Once the data is clean, answer these business questions.

The Client asks:

1. **The "Golden Hours":** "When is our fleet busiest? I want a heatmap or bar chart showing the average number of trips by **Hour of Day**. Do we need more drivers at 5 AM or 5 PM?"
2. **The Solo vs. Group Traveler:** "Do people mostly travel alone or in groups? What is the most common passenger_count? Does the tip_amount increase if there are more passengers?"
3. **The Payment Split:** "What percentage of our customers pay by Credit Card vs. Cash (payment_type)? Is there a difference in the average tip_amount between card users and cash users?" (*Hint: This is a classic behavioural insight.*)
4. **The Airport Run:** "We suspect trips with RatecodeID = 2 (JFK Airport) have a higher average fare than standard city trips (RatecodeID = 1). Can you prove this with data?"

Phase 3: The Strategic "So What?" (Storytelling)

Don't just give me a table. Give me a recommendation.

The Client asks:

1. **Congestion Impact:** "We charge a congestion_surcharge. Does this surcharge discourage people? Compare the trip volume for trips *with* surcharge vs. *without*. What is your recommendation? Should we lobby to remove it?"
2. **The "Whale" Locations:** "Which Pickup Location (PULocationID) generates the highest **total revenue**? Identify the Top 3 locations and recommend where we should station our empty taxis."

Phase 4: The New "Must-Haves" (Advanced Data Engineering)

The Client asks:

1. **The "Time Travel" Bug (Duration Calculation):**
   - "We need a trip_duration column (in minutes). But be careful—I ran a quick check and found trips with **negative duration** (e.g., Dropoff was *before* Pickup). Find them, count them, and kill them. They are corrupting our averages."
2. **The "Rocket Car" Anomaly (Speed Analysis):**
   - "Calculate the average speed (speed_mph) for every trip.
   - **Logic Check:** Speed = Distance / (Duration_in_Hours).
   - **The Problem:** We suspect some drivers are spoofing GPS. If a trip shows a speed > 100 MPH or < 0 MPH, flag it as 'Data Error'. How many rows do we lose?"

Phase 5. The "Vendor Wars" (comparative Analysis)

The Client asks:

1. **VeriFone (1) vs. Creative Mobile (2):**
   - "We use two providers for our in-car systems: Vendor 1 and Vendor 2.
   - Vendor 2 handles ~70% of our rides. But who is actually better?
   - **The Challenge:** Compare the **Average Tip Amount** for Vendor 1 vs. Vendor 2. Does the interface of one machine encourage better tipping? This is a huge potential revenue find for our drivers."

Phase 6. The "Mystery" Codes (Data Detective Work)

The Client asks:

1. **The "RateCode 99" Mystery:**
   - "Standard trips are RateCode 1. JFK Airport is RateCode 2.
   - But my tech team says we have 64 trips with **RatecodeID = 99**.
   - What are these? Are they high value? Low value? Scams? Isolate these 64 rows and tell me what they have in common."

Phase 7. The "Route Master" (Geospatial Logic)

The Client asks:

1. **The "Commuter Loop":**
   - "Find the Top 5 most popular **pairs** of Pickup (PULocationID) and Dropoff (DOLocationID) locations.
   - *Hint:* Create a new column route = PULocationID -> DOLocationID.
   - **The Question:** Are the top routes long cross-city trips, or just short hops (e.g., 237 -> 236)? If they are short hops, we should deploy electric scooters there instead of taxis."

Technical Requirements (The Deliverables)

To accept your work, I need:

1. **A Python Notebook (.ipynb)** showing all your cleaning steps. (Show your work!)
2. **A cleaned_taxi_data.csv** file (The final, trusted dataset).
3. **A 7-slide Executive Deck** answering the questions in Phase 2 to 7 with clear charts.

Good luck. The Board is waiting.